Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

CRITICAL CARE

Organ Donation After Circulatory Death (DCD)

YG CG 1015 May 2022

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Key Points

- This guideline should be taken as advice for the theatre management of donors, once care has been optimised by and handed over from the Intensive Care team, as per national guidelines and in liaison with the Specialist Nurse for Organ Donation (SNOD).
- Also refer to YG CG 1014: Organ Donation after Brain Death (DBD)

General Principles:

- Preparation and timing is key
- An anaesthetist/intensive care doctor is required to oversee withdrawal of care and confirm death only. In the event of DCD lung donation, anaesthetic support will however be required.
- The SNOD will coordinate the arrival and briefing of the various teams, book the patient on the theatre list and be the primary point of contact with the family
- At YG, retrieval will not normally occur in Theatre 4, but in an unoccupied theatre, to be discussed between the Consultant Anaesthetist on call, Theatre Managers and Co-ordinator (Bleep 611).

Anaesthetic Considerations:

- **Initial Responsibilities:**
  - Discussion between ITU team and anaesthetist to clarify the location of treatment withdrawal and the role of the anaesthetist
  - Take over care from the ITU team and transfer the donor, if withdrawal is to take place in theatre. This may involve changing over to anaesthetic room/theatre monitoring.
  - Provide blood sampling if requested
- **Position and access:**
  - Supine, with arms at side for surgical access
  - Access would be required for palliative anticipatory medications, not for the purposes of retrieval
- **Anaesthesia:**
  - The patient will not be ventilated once on the operating table except if lungs are to be retrieved (see below)

DCD excluding lungs donation:

- Treatment withdrawn in ITU or in Anaesthetic room. This is to be discussed with ITU team and SNOD.
- Family can be present during this time and are best supported by the additional presence of the SNOD
- The organ retrieval can only go ahead if asystole occurs within 3 hours of treatment withdrawal (SNOD to monitor time)
- Death is confirmed during the subsequent 5 minutes of asystole by an anaesthetist (identified in advance) using the form “The Diagnosis of Death following Cardiorespiratory Arrest”- printable from the hospital intranet or as part of ICU End of Life Pathway Paperwork. An anaesthetist should remain with or near the patient to proceed with confirmation of death as soon as possible after asystole.
- Immediately after confirmation of death the patient is transferred to operating room (from anaesthetic room or directly from ITU).
- Once the donor is safely transferred onto the operating table, the YG anaesthetist is no longer required but may choose to remain.

DCD including lungs donation:

- The SNOD will provide a standard NHSBT lung donation checklist which helps track timings
- As above for treatment withdrawal and confirmation of death
- Patient will be transferred to operating room if not already there
- Once on the operating table, the patient needs to be re-intubated (to protect the lungs from soiling). DO NOT RE-INFLATE THE LUNGS at this point.
- THE LUNGS WILL BE RE-INFLATED at 10mins from onset of irreversible asystole (the SNOD will keep track of the time), with a single vital capacity breath.

  This is achieved by a recruitment manoeuvre,

  - generating an airway pressure of 40cmH20 for 30 seconds,
  - followed by a maintenance of 5 cmH20 using the APL valve, OR simply by clamping the ETT.
  - DO NOT ventilate after this single breath
- The retrieval team may later ask for cyclical ventilation to be restarted, but this is only done after cold perfusion of the lungs and venting of the left atrium. The Cardiothoracic surgeons will guide the local anaesthetist with the specifics.

Donors	Local Anaesthetic Team	Retrieval cardiothoracic Team
DBD **With heart/lung donation**	Transfer patient to theatre. IPPV and circulatory support until cross clamp. Consider positioning, ventilation, anaesthesia, steroid/antibiotics, heparin, blood sampling.	Assessment and management Includes PAC, TOE, Bronchoscopy, inotrope & hormone therapy
DBD **No heart/lung donation**	Transfer patient to theatre. IPPV and circulatory support until cross clamp. Consider positioning, ventilation, anaesthesia, steroid/antibiotics, heparin, blood sampling.	Not present
DCD **With lung donation**	Transfer patient to theatre. Withdraw support in anaesthetic room. Prescribe/administer PRN. Confirm death. Re-intubate and re-inflate lungs. Transfer to theatre.	Not present
DCD **Without lung donation**	Transfer patient to theatre. Withdraw support in anaesthetic room. Prescribe/administer PRN. Confirm death. Transfer to theatre.	Not present

References

- Whittington Health Organ Donation Guideline, 2013
- Management of the Heart-beating Brain Dead Organ Donor, British Journal of Anaesthesia 108 (S1): i96–i107 (2012)
- Management of the Potential Heart-Beating Organ Donor, CEACCP Volume 4 Number 3 2004